



Estimating a Game Theoretic Model

WIETZE LISE

*Institute for Environmental Studies, Vrije Universiteit Amsterdam, De Boelelaan 1115,
1081 HV Amsterdam, The Netherlands, E-mail: wietze.lise@ivm.vu.nl*

(Accepted 27 January 2000)

Abstract. Applied game theory generally shows possible game situations, discussing their implications. This paper derives an estimation procedure for non-cooperative games, using cross-section data. During the transformation of the data an attempt is made to pick extreme points, to find the outer borders for the actual game. The estimation procedure is an algorithm that switches between several software packages. In an application a motivated choice can be made for the transformation functions such that the estimated game comes closest to the actual game of the empirical problem under study. To demonstrate the estimation procedure, a situation is studied in which a villager voluntarily chooses a level of participation in forest management.

Key words: algorithm, cross-section data, estimation procedure, noncooperative games, two-person symmetric games

List of Symbols

α	constant	N	set of all interviewed members in all organizations together
β	regression coefficient	N_j	set of interviewed members in organization j
$\mathcal{R}$	set of real numbers	o	number of organizations
ϵ	error term	O	set of organizations
Γ	participation game	p_{ij}	price of product j accruing to player i
λ_j	transformation function	q_{ij}	quantity of product j accruing to player i
μ	transformation function	r	number of goods which benefit a player
$\bar{\vartheta}_i$	transformation by taking the mean	s_i	arbitrary variable
π_i	payoff to player i	x, y, a, b	payoffs to player i
σ_{ϑ_i}	transformation by taking the variance	X, Y, A, B	payoff groups containing all observations related to payoffs x, y, a, b
θ^*	threshold value of participation	$ X $	cardinality (number of observations) of set X
θ_i	action of player i	D	do not participate
i	challenger	$\bar{D}$	set of non-participators
$-i$	contender	P	participate
j, k	index for a good which can benefit the player, index for participatory indicator, or index for organization	$\bar{P}$	set of participators
m	number of participatory indicators	PI_i^j	the j th participatory indicator for person i
n	or number of players in a game	$\mathbf{PI}_i$	vector of PI_i^j for person i
		$\mathbf{SE}_i$	vector of socio-economic variables for person i

1. Introduction

Applied game theory has grown in popularity. The values of the payoffs, however, are generally selected illustratively. This paper proposes a stepwise procedure for deriving the payoffs and actions in a game from (primary) cross-section data, in order to introduce more rigor in deriving the payoff matrix.¹ Hence, this procedure starts from a diametrically different viewpoint: the game is assumed not to be known with certainty in a practical situation, but the data will tell which game is actually being played. A common economist's assumption is relaxed here to enable such an estimation. We assume that the society is not in equilibrium. This gets closer to the reality than mainstream economics, on which Coase comments: "What is studied is a system which lives in the minds of economists but not on earth" [2].

This paper considers a game theoretic model of a situation in which members of a particular organization are free to choose whether or not they want to participate in the activities of the organization. Examples of such organizations are a fishery community, a society for forest management, a firm, etc. An estimation procedure is derived for the game theoretic model. This paper focuses on the technical method for estimating a game. The difficulty in estimating such a game is to find the payoffs of actions which are not directly observable, but which could have been taken. To this purpose an algorithm is proposed which switches between several software packages to ultimately estimate the game, viz., a programming package like Turbo Pascal, an econometric package like SPSS, and a package handling spreadsheets like Excel. Observations on members across a number of organizations, about their payoffs and actions are required in the form of cross-section data. An important premise is that we do not know exactly in advance which game these members of the organization are playing. Whether they are in equilibrium or not is not known, either.

This paper is organized as follows. The choice situation of people's participation in an organization is formalized in Section 2. A taxonomy of two-person symmetric games is derived to enlist all possible choice situations. Section 3 proposes the estimation procedure for a non-cooperative game. Subsection 3.1 deals with a method to measure the basic ingredients of a noncooperative game: payoffs and actions. Indicators of participation are used to derive the action of the interviewed person, the *challenger*, and the joint action of *other* interviewed persons in the same organization, the *contender* (Subsection 3.2). Subsection 3.3 presents two methods for deriving the payoff matrix: the Euclidean Cluster Method and the Homogeneous Grouping Method. Section 4 applies the estimation procedure. Data are used from a study in Haryana, a state of India. Section 5 discusses the possibility of using similar techniques for more complicated games and other situations as a scope for future research.

Table I. The payoff-matrix of participation game Γ .

Challenger	Contender	
	Participate (=P)	Do not participate (=D)
Participate (=P)	x, x	b, a
Do not participate (=D)	a, b	y, y

2. A Game Theoretic Model

It is a puzzling question why some people are more actively involved in an organization than others. To explain differences in levels of participation, let us model such a situation as a game in which members of a particular organization adopt a level of participation (= choice), while the organization provides them additional goods (= their payoff). We refer to this game as the *participation game*. The simplest form of such a game consists of only two persons, who have a choice between two alternatives: to participate or not. Let us assume that both persons are identical. In general, members differ in income, power and so on, but they face the same choice situation. The assumption is plausible, as long as the differences between the members are not too large.

The participation game can be formalized as follows.

- The set of players consists of two persons: $\{1, 2\}$. Let us label the two persons as challenger i and contender $-i$.
- Player i takes action θ_i . Abbreviate the set of actions {Participate, Do not participate} as $\{P, D\}$.
- The payoff to the challenger i is $\pi_i(\theta_i, \theta_{-i})$, where θ_i is the action taken by player i . The total payoff set in this situation consists of four different numerical values: $\{x, y, a, b\}$.

In this way, the participation game is formalized as a two-person symmetric game Γ . Table I shows the payoff-matrix.

A special feature of this game is that there are no restrictions on the values of the payoffs. Given the derived values of the payoffs, a particular situation emerges representing either a situation of harmony or conflict. Comparison of this outcome with the actual situation in a practical situation can provide insight into the rigor of the estimation technique.

Let us assign the following important economic interpretation to a main difference in game Γ .

DEFINITION 2.1. $x - y = \text{the net incentive to participate.}$

If the net incentive to participate $x - y$ is positive, we are dealing with a meaningful game situation, where mutual participation is more beneficial than mutual non-

participation. However, in practical situations it is not guaranteed that $x > y$. Refer to this as the *reverse* situation.

DEFINITION 2.2. *If $y > x$, the net incentive to participate is negative, we are dealing with a reverse game.*

It is useful to classify all possible games. For that, we have to consider all 24 (that is 4!) possibilities for ordering the payoffs a, b, x, y . These can be divided broadly into six distinct cases. Let us concentrate on strict inequalities. We call games with equal payoffs *transition games*.

In 12 out of 24 possible cases, either to participate or to do not participate is a dominating action.

DEFINITION 2.3. *An action is dominating when it is the best response to all possible actions of the opponent.*

Then the game has a unique Nash equilibrium, which is the combination of the two dominating actions.

DEFINITION 2.4. *An action $\hat{\theta}$ is said to be a Nash equilibrium for Γ , if the following inequalities are satisfied.*

$$\pi_i(\hat{\theta}) \geq \pi_i(\theta_i, \hat{\theta}_{-i}) \text{ for all } i \in \{1, 2\} \text{ and for all } \theta_i.$$

In 10 of the 12 cases the individual best response matches with the collective preferred outcome. We call this the *Pareto game*. In the remaining two cases, the individual best response conflicts with the collective preferred outcome. This is the well-known prisoner's dilemma.

In the other 12 cases there is no dominating action, but there are always two pure-strategy Nash equilibria. In 6 of these 12 cases, mutual participation is one of the Nash equilibria of the game. Then it is purely a matter of coordination to arrive at this equilibrium. These games are known as coordination and assurance games. However, in the other 6 cases, the Nash equilibria consist of a mix of participators and non-participators. Then it is optimal for players to take opposite actions; to non-coordinate their actions. These games will be called non-coordination games and the well-known chicken game. Even in the non-coordination game, actions should be coordinated, but in such a way that players take opposite actions.

The coordination and non-coordination games are very similar to the assurance and chicken games, respectively. There is, however, a difference. For the (non-) coordination game the Nash equilibria payoffs are strictly greater than the remaining payoffs, which is not true for the assurance and chicken games. Table II summarizes the derived complete taxonomy of two-person two-action symmetric games.

Table II. Classification of games based on the payoff ordering.

Name of the game	Payoff ordering	
	Standard	Reverse
1a prisoner's dilemma	$a > x > y > b$	$b > y > x > a$
1b Pareto game	$x > \max\{a, y\}; b > y$	$y > \max\{b, x\}; a > x$
2a assurance game	$x > a > y > b$	$y > b > x > a$
2b coordination game	$x > y > \max\{a, b\}$	$y > x > \max\{a, b\}$
3a chicken's game	$a > x > b > y$	$b > y > a > x$
3b non-coordination game	$\min\{a, b\} > x > y$	$\min\{a, b\} > y > x$

3. Estimation Procedure

3.1. QUANTIFYING PAYOFFS AND ACTIONS

3.1.1. *The Amount of Incremental Income*

Quantifying the amount of incremental income that flows to a member of an organization can best be illustrated by an example. In the case of forest management, such goods as fuelwood, fodder, irrigation water, grazing rights, goodwill of other people, ease in collecting resources, etc. are all gains from participation. The exact gain varies from situation to situation, depending on the particular institutional structure and how benefits are distributed.

Let the amount of incremental income π_i be the total sum of the imputed value p_{ij} of the products q_{ij} which accrue to members of a particular organization.²

$$\pi_i = \sum_{j=1}^r p_{ij} q_{ij} . \quad (1)$$

Adding all such separate components gives the total gain from an organization to a particular member: the payoff.

DEFINITION 3.1. π_i is the incremental income accruing to person i due to the presence of a local organization.

3.1.2. *Indicators of Participation*

Various situations exist in which a person can decide whether to participate or not. Some institutions encourage participation, while other institutions can do the opposite.

The choice situation which is of interest here is the extent of people's participation in a particular organization. The basic postulate is that the choice to participate or not is voluntary. To measure participation, it is not sufficient to know whether

or not a person is a member of an organization: the choice to become a member is made only once; it does not account for changes in perception during the process. Some people can be involved very actively, while others only reap the benefits. To separate such kinds of people, it is necessary to look for alternatives in quantifying participation.

Consider the following components of people's participation.³

- (i) Contributing.
- (ii) Sharing in the benefits.
- (iii) Involvement in decision-making and evaluation.

People can choose whether or not to participate in many aspects of an organization. The aspects mentioned above are three such choice situations. Participation means taking part in something.

This distinction can be used to pose questions about a person's participatory attitude. Answers to these questions can, for example, be recorded on an ordinal scale, as follows:

Not at all _____ little bit _____ medium _____ quite a lot _____ very much
 1 _____ 2 _____ 3 _____ 4 _____ 5

The answers to each question can be framed so that a higher number represents more participation. This way of collecting a response from a member of an organization is used to specify the actual level of participation.

DEFINITION 3.2. Let PI_i^j represent the reply of person i to question j as the j th participatory indicator of person i .

Denote vector $(PI_i^1, PI_i^2, \dots, PI_i^m)$ as $\mathbf{PI}_i$. In the remainder of this paper, bold variables stand for vectors. In our present example, a participatory indicator can take any integer value between 1 and 5: $PI_i^j \in \{1, 2, 3, 4, 5\}$.

3.2. DERIVING THE LEVEL OF PARTICIPATION

3.2.1. Choice of the Challenger

The participatory indicators $\mathbf{PI}_i$ of Section 3.1.2 represent a set of responses to a variable, which is not directly observable: participation. Denote the level of participation for individual i as θ_i . In general, θ_i is a function of $\mathbf{PI}_i$. Let us denote this function as μ , which converts m indicators of participation into a single value:

$$\begin{aligned} \mu : \mathcal{R}^m &\rightarrow \mathcal{R}; \\ \theta_i = \mu(\mathbf{PI}_i) &= \mu(PI_i^1, PI_i^2, \dots, PI_i^m), \end{aligned} \tag{2}$$

where m is the number of relevant questions asked during the survey and $\mathcal{R}$ is the set of real numbers.

The simplest way to define the level of participation is by taking the grand total of all participatory indicators $\mathbf{PI}_i$ as defined in Equation (3).

$$\theta_i = \sum_{j=1}^m \mathbf{PI}_i^j. \quad (3)$$

Let θ_i^{ts} denote the level of participation based on the total sum of the indicators of participation.

An alternative way of defining the level of participation is by undertaking a factor analysis [4–6] on the indicators of participation. This is a method for translating a large set of variables into a few choice-variables: the principal components.

It is possible that the set of participatory indicators results in more than one principal component. In general, as a rule of thumb, all principal components with a variance greater than 1 are used for further analysis. When the variance in the second or even the third principal component is greater than one, participation is multidimensional. Another rule of thumb is that the indicators with a factor loading more than 0.5 in absolute value are dominant.

Since the principal components are orthogonal: each principal component reflects a different choice situation for participation. Let $\theta_i^{\text{pc}_k}$ denote the level of participation based on the k th principal component of the indicators of participation.

3.2.2. Choice of the Contender

Consider the following formalization, required in order to derive the choice of the contender. Define the set of organizations as $O := \{1, 2, \dots, o\}$ in which a particular organization exists. Let N_j , for $j \in O$, be the set of $|N_j|$ (surveyed) members in organization j , where $|N_j|$ is the cardinality of set N_j . Let $N := \bigcup_{j \in O} N_j$ be the collection of all observations and let $|N|$ be the total number of people surveyed in all organizations together.

Consider the following payoff function:

$$\pi_i(\theta_i, \boldsymbol{\theta}_{-i}),$$

where θ_i is the level of participation of challenger i belonging to organization j . The collective choice of all other members $\boldsymbol{\theta}_{-i}$ is an $(|N_j| - 1)$ -vector representing all the individual levels of participation in organization j , excluding person i .

$$\boldsymbol{\theta}_{-i} := (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_{|N_j|}) \text{ where } i \in N_j. \quad (4)$$

In the notation introduced so far, the model is very general, where the decision of a single player affects the payoff of other players. Then the players are known to act *strategically*. To fit the $|N_j|$ -person game at the organizational level into the participation game, assume that each member is comparable, irrespective of wealth, caste or creed. This assumption also implies the symmetry of the game.

Irrespective of the sample size, each particular organization can be converted into a two-person game by introducing an opponent: the contender.

In an empirical setting, however, it seems likely that people act more impulsively than strategically. When members act impulsively, they observe the outcome of the actions of all other members. It is less likely that they deduct each and every action of all other members. Contrast this with strategic action, where the actions of each individual can be observed. It seems plausible to assume that a person's choice is conditioned as if all other members had acted jointly. In this case, the other's actions are taken as a simultaneous move.

By summarizing the $(|N_j| - 1)$ -vector into a scalar, the situation can be analyzed as a two-person game. Let λ_j be the transformation function which converts vector θ_{-i} into a single value.⁴

$$\begin{aligned} \lambda_j : \mathcal{R}^{(|N_j|-1)} &\rightarrow \mathcal{R}; \\ \vartheta_i &:= \lambda_j(\theta_{-i}) \text{ for all } i \in N_j \text{ and } j \in O. \end{aligned} \quad (5)$$

Function λ_j converts the actions of $|N_j| - 1$ other members in organization j into a *composite* second person. We assume here that each member of an organization treats all others as composite second players. The composite second player is the contender who takes action ϑ_i . In this manner, the game retains the flavor of an n -person game. The payoff function can now be written as $\pi(\theta_i, \vartheta_i)$. Consider the following two functional forms of λ_j :

- I Mean: $\lambda_j(\theta_{-i}) := \bar{\vartheta}_i$.
- II Variance: $\lambda_j(\theta_{-i}) := \sigma_{\vartheta_i}$.

Many other functional forms for $\lambda_j(\theta_{-i})$ are possible.⁵ These are of interest because of their simplicity and complementarity.

Equation (6) uses the notation introduced so far to present the mean level of participation in organization j .

$$\bar{\theta}_j = \frac{1}{|N_j|} \sum_{i \in N_j} \theta_i \text{ for all } j \in O. \quad (6)$$

Since we are looking at the average participatory behaviour of the other persons in a certain organization, as opposed to each person i with a participation level θ_i , it is obvious to rely only on *their* mean level of participation, by excluding the level of participation of person i . This leads to the functional form of $\lambda_j(\theta_{-i})$ represented in Equation (7).

$$\bar{\vartheta}_i = \frac{1}{|N_j| - 1} \sum_{\{k \in N_j \mid i \neq k\}} \theta_k \text{ for all } i \in N_j \text{ and } j \in O. \quad (7)$$

When ϑ_i is based on the mean, we call it model I. It is denoted by putting a bar ($\bar{}$), above the variable. This situation can be interpreted as action risk-neutral, because

all actions of other members are weighed equally by taking the mean. Let us refer to model I as the *mean situation*.

When we take the variance of the levels of participation of all other members of a particular organization as the action of the contender, the situation is intrinsically different. In contrast to the mean situation where all deviations are accounted for equally, the ‘variance’ situation is characterized by squares of the variation from the mean of the actual level of participation. In this way, people with the highest deviation from the mean are accounted for much more than the people who are close to the mean. Positive and negative deviations are no longer cancelling each other out. Every deviation is added to the total value of the action of the contender. Equation (8) represents the ‘variance’ situation.

$$\sigma_{\vartheta_i} = \frac{1}{|N_j| - 1} \sum_{\{k \in N_j | i \neq k\}} (\theta_k - \bar{\theta}_j)^2 \text{ for all } i \in N_j \text{ and } j \in O. \quad (8)$$

When ϑ_i is based on the variance we call it model II. It is denoted by putting the variable as a subscript to σ . This situation can be interpreted as action risk-averse. Let us refer to model II as the *variance situation*.

3.3. TWO GROUPING METHODS

3.3.1. Euclidean Cluster Method

We propose two methods to derive the payoff-matrix by dividing the observations into four groups. In the first method, we need normalized levels of participation of both the challenger and the contender in order to eliminate a bias either towards the contender or the challenger. In general, variable s_i can be converted into a fraction between 0 and 1 by applying Equation (9).

$$\tilde{s}_i = \frac{s_i - \min_{k \in N} s_k}{\max_{k \in N} s_k - \min_{k \in N} s_k} \text{ for all } i \in N, \quad (9)$$

where the symbol ($\sim$) is put above a variable to denote that it is normalized between zero and one.

The members are assigned into four clusters, so that the Euclidean distance within clusters is minimized, while the Euclidean distance between clusters is maximized.⁶ Call this way of dividing the observations the Euclidean Cluster Method. The Euclidean Cluster Method also yields final cluster centers. Table III shows how to assign the final cluster centers of the actions of the challenger (θ) and the contender (ϑ) to payoff-groups A , B , X , Y . Section 4 gives an example of the Euclidean Cluster Method.

Table III distinguishes between the mean and the variance situation, because they require a different interpretation. In the mean situation, a large value of a cluster centre always stands for ‘participation’, while a small value means ‘no participation’. In the variance situation, the action of the contender is conditional

Table III. Assigning members into clusters, using final cluster centers.

Mean situation			Variance situation		
θ	ϑ	Payoff group	θ	σ_{ϑ}	Payoff group
'large'	'large'	x (P,P)	'large'	'large'	b (P,D)
'large'	'small'	b (P,D)	'large'	'small'	x (P,P)
'small'	'large'	a (D,P)	'small'	'large'	a (D,P)
'small'	'small'	y (D,D)	'small'	'small'	y (D,D)

Note: θ , $\bar{\vartheta}$, σ_{ϑ} : levels of participation of the challenger (θ) and the contender (ϑ). P = participate, D = do not participate.

on the action of the challenger. When the variance is small, the actions of the challenger and the contender are the same, but when the variance is large, the action of the contender must be opposite to the action of the challenger.

The payoff-matrix of Table I can be computed by taking average payoffs of the assigned observations to each cluster. Formally, the payoff can be calculated by applying the following formula, where $|A|$ denotes the number of observations in payoff group A:

$$a = \frac{1}{|A|} \sum_{i \in A} \pi_i; b = \frac{1}{|B|} \sum_{i \in B} \pi_i; x = \frac{1}{|X|} \sum_{i \in X} \pi_i; y = \frac{1}{|Y|} \sum_{i \in Y} \pi_i. \quad (10)$$

3.3.2. Homogeneous Grouping Method

Another way to derive a payoff matrix is by assigning the observations into four *equal* groups. This can be done in two steps.⁷

The first step separates participators and non-participators, using the level of participation θ_i . We take the *median* of the level of participation θ_i as the separating threshold value θ^* .⁸

All members who have a level of participation at least as large as the threshold value θ^* are assumed to participate, while others do not participate. In general, k members participate and $n - k$ members do not participate. The separation between participators and the non-participators can be denoted formally as follows.

$$\begin{aligned} P &= \{i \in N | \theta_i \geq \theta^* \text{ (person } i \text{ is participating)}\} \\ D &= \{i \in N | \theta_i < \theta^* \text{ (person } i \text{ is not participating)}\}. \end{aligned}$$

Note that $P \cap D = \emptyset$ and $P \cup D = N$ represent all persons in the sample.

The second step splits these two sets into two more subsets by taking the *median* of the subset P or D as the threshold value.⁹ Denote these as ϑ_P^* and ϑ_D^* respectively. In this manner, it is possible to realize four subsets of observa-

tions combining large and small levels of participation. Equation (11) specifies the required division.

	Mean	Variance	
$\theta_i \geq \theta^*, \vartheta_i \geq \vartheta_P^*$ (large, large)	$\Rightarrow i \in X$	$i \in A$	
$\theta_i \geq \theta^*, \vartheta_i < \vartheta_P^*$ (large, small)	$\Rightarrow i \in A$	$i \in X$	(11)
$\theta_i < \theta^*, \vartheta_i \geq \vartheta_D^*$ (small, large)	$\Rightarrow i \in B$	$i \in B$	
$\theta_i < \theta^*, \vartheta_i < \vartheta_D^*$ (small, small)	$\Rightarrow i \in Y$	$i \in Y$	

Consult Table III for assigning the observations into payoff groups. Call this way of dividing the observations the Homogeneous Grouping Method.

As before, the payoff-matrix (Table I) can be derived by calculating the average payoffs of all observations in each group. Further, it is also possible to base the payoffs on several socio-economic variables, within each payoff-group. This can be done by estimating the following regression equations:

$$\pi_i = \beta^T \mathbf{SE}_i + \alpha + \epsilon; \quad i \in A, B, X \text{ or } Y, \quad (12)$$

where $\mathbf{SE}$ is a vector of socio-economic variables, β and α the regression coefficients, ϵ the error term. This results four times in a regression on subsets of the total data set using the division of Equation (11).

Estimation of Equation (12) for each subset of data proceeds in two steps. Step one derives which socio-economic variables should be included in the regression Equation (12).¹⁰ Step two regresses the final equation, which can be used for further analysis. In an alternative method relevant socio-economic variables can be selected irrespective of the resulting significance level. Such a method, although less reliable, is tailor-made for analyzing the impact of certain policies.

3.3.3. Comparison

The Euclidean Cluster Method gathers the observations around the equilibrium of the game. In some cases, the equilibria of the game can be found even without calculating the average payoffs. This simplifies the analysis considerably.

In the Euclidean Cluster Method, four clusters must be calculated. This means that at least one observation is assigned to each group. It sometimes happens that the final group centers are not statistically different.¹¹ Most of the time, it is possible to calculate four different groups meaningfully.

The Euclidean Cluster Method is hard to use for a regression in each cluster, because the number of observations can be very unevenly distributed. The extreme possibility exists of having one observation in a cluster. This single observation solely determines the final payoff. This problem can be avoided by forcing the observations into certain groups. The Homogeneous Grouping Method is designed for that purpose.

The Homogeneous Grouping Method forces the observations into four groups of the same size. The Homogeneous Grouping Method focuses on the relative

Questionnaire:

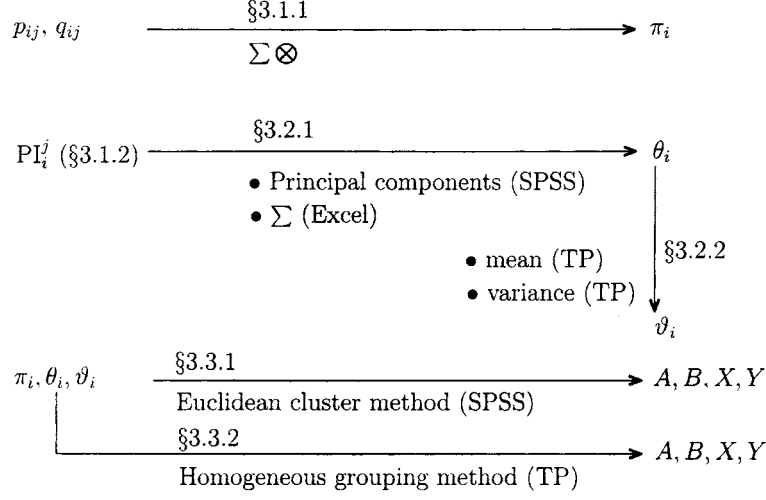


Figure 1. A flow chart of the estimation procedure.

differences in the payoffs. It does not consider how close the actions of players are together. This makes the Homogeneous Grouping Method less useful for interpreting the game at the institutional level. This is why the Euclidean Cluster Method is preferred to the Homogeneous Grouping Method from a game theoretic point of view. For analyzes at the organizational level, the Homogeneous Grouping Method is the best we can do from a statistical point of view.¹²

Figure 1 summarizes the algorithm which we derived in this section for estimating a game theoretic model.

4. An Example of Estimating Payoff Functions

This section shows how to estimate a payoff matrix in a practical situation. Data on a village council in Haryana state of India is used in the examples for applying the Euclidean Cluster Method and the Homogeneous Grouping Method [8, 9].

In order to apply the algorithm, we need to choose appropriate functions for μ and λ_j . Let the action of the challenger be based on the total sum of the participatory indicators (μ) and the action of the contender be based on the mean level of participation of all other interviewed people in a village (λ_j). Table IV shows how the payoffs are assigned to a particular cluster by considering the final cluster centers. The largest cluster centers are underlined to show the meaning of ‘large’ and ‘small’ in Table III. For instance, the combination of the two largest cluster centers corresponds to payoff-group X. The other clusters are assigned to their payoff groups similarly. The number of observations in each cluster and the average payoffs are displayed as well.

Table IV. An example of the Euclidean Cluster Method.

Cluster	θ^{ts}	$\bar{\theta}^{ts}$	π	Payoff	Cases
4	<u>0.8236</u>	<u>0.8287</u>	x	3766	55
2	<u>0.6777</u>	0.3553	b	5366	53
1	0.2804	<u>0.6406</u>	a	3528	8
3	0.2555	0.1533	y	1583	11

The same situation is also estimated with the Homogeneous Grouping method. Table V shows the result. The levels of participation in the payoff-groups can also be compared with each other. This can be done by running the MEANS-T-TEST in SPSS (Section 3.3.3). It turns out that all differences are significant with at least 95% confidence.

As Section 3.3.2 argues, it is also possible to base the payoffs on several socio-economic variables, with a significance of all included variables above 95%. The values in the brackets denote the averages of the included socio-economic variables, within the payoff-group.

$$\begin{aligned}
 x &= +6.93 (32.5) \text{ FODTOT} + 3836 (0.830) \text{ FORDEP} + 0.0110 (24606) \\
 &\quad \text{INCOME} + 113 (15.0) \text{ FWTOT} + 2.40 (793) \text{ BAMTOT} - 3117; \\
 b &= -0.00194 (1529952) \text{ CAPITAL} + 305 (30.7) \text{ LANQU} : \\
 a &= +97.6 (16.2) \text{ FWTOT} + 227 (6.23) \text{ ANIMUN}; \\
 y &= +5023 (0.670) \text{ FORDEP}
 \end{aligned}$$

The included variables have the following meaning:

FODTOT	=	Total use of fodder. [quintals ¹³ per year]
FORDEP	=	Forest dependence [fraction of the amount of goods collected from the forest and the total need of those goods].
FWTOT	=	Total use of fuelwood. [quintals per year]
BAMTOT	=	Total use of bamboo. [pieces per year]
LANQU	=	Total output of land. [quintals per year]
ANIMUN	=	Animal unit. [cow-units]

Let us concentrate on the total use of fuelwood, by substituting the average values of all other factors, to see the impact of changes in socio-economic variables on the payoff structure. The set of payoff-equations can be simplified as follows:

Table V. An example of the Homogeneous Grouping Method.

Group	θ^{ts}	$\bar{\theta}^{ts}$	π	Payoff	Cases
1	<u>0.8626</u>	<u>0.9099</u>	x	4151	36
2	<u>0.7969</u>	0.5584	b	6118	33
3	0.5070	<u>0.6053</u>	a	3135	29
4	0.4902	0.2020	y	3273	29

$$\begin{aligned}
 x &= 2466 + 113 \text{ (15.0) FWTOT;} \\
 b &= 6395; \\
 a &= 1414 + 97.6 \text{ (16.2) FWTOT;} \\
 y &= 3365.
 \end{aligned}$$

A decrease in the total need for fuelwood from 15 quintals per year to 7.9 leads to a structural change in the payoff structure. The Pareto game then becomes a reverse prisoner's dilemma. Hence, whenever the payoffs become equal, due to changes in the socio-economic variables, a point of transition is reached. This leads to a structural change when the socio-economic changes are further intensified.

By estimating payoff functions, a sensitivity analysis is possible on the payoff structure and a game with varying payoffs can be derived. Variation in the values for these socio-economic variables anticipates a dynamic process.

5. Conclusions and Extensions

This paper has derived an algorithm for estimating a game theoretic model, based on cross-section data. The algorithm is applied to the special case of people's participation in forest management.

Possibilities to quantify two basic ingredients of a game, action and payoff, may be pointed out. The level of participation is based on several participatory indicators and the payoff is based on the imputed value of the additional products from a particular organization. The action of the challenger is taken as the grand total or as the first or second principal component of the indicators of participation. The action of the contender is based on the mean or the variance of the levels of participation of all other surveyed members of a particular organization.

The complete data-set is divided into four groups based on the level of participation of both the challenger and the contender. This is done through the Euclidean Cluster Method or the Homogeneous Grouping Method. In the Homogeneous Grouping Method, as the name suggests, one tries to divide the observations into groups of the same size. The payoff-matrix is derived by taking the average values of the payoffs within each group. The four payoffs of the game do not need to remain fixed, but can be based on socio-economic variables. Changes in these

variables can induce changes in the relative payoff-structure. This paper considers a number of organizations for estimating the ‘average game at the organizational level’.

There are possibilities to extend the model to higher dimensions. Extension to a repeated game is possible by simply assuming that the derived one-period game is played repeatedly. A further extension is possible by using the Homogeneous Grouping Method. By basing the payoffs on several socio-economic variables, it is possible to formulate a repeated game with changing payoffs as well. Unfortunately, the general analytical solutions for such games are complex.

Extension of the model to a continuous choice framework seems possible with the current data set. We should then work with non-linear payoff functions. The first-order conditions provide the Cournot reaction functions to see whether a benchmark solutions is possible as a joint utility maximization outcome [13].

Extending the model of participation to n -persons leads to technical difficulties. For estimating a game with more than two persons, the varying number of $|N_j|$ members has to be converted into a fixed number of n members for any desired number n .

Extension to more than two persons, however, is only possible for the mean situation. In the variance situation, if one player participates and a second player does not, a high variance is not conclusive for the action of the hypothetical third player. This shows the limitations of extending the variance situation to more than two players.

An area for further research is to extend this model to a two-person asymmetric game. Extension is obtained by allowing for different payoff structures π_i and π_j when $i \neq j$. For example, when person i is facing a prisoner’s dilemma it is quite possible that person j is facing an assurance game, $i \neq j$. Consider, for example, the game between the state and the villagers. The priorities for the state can be quite different from the priorities of the villagers. Then they do not need to face the same game situation and the assumption on the similarity of the players is no longer valid. For instance, in the situation where the state faces a prisoner’s dilemma and the villagers face an assurance game, the Nash equilibrium consists of a non-participating state, while the villagers are better off by continued participation. To estimate the asymmetric game, data should first be divided into two groups, i.e., in the last example they are (1) the workers of the state and (2) the villagers. These two groups can be analyzed in isolation as before, to derive the game situation they are facing. A taxonomy of ten game situations for the case of asymmetric two-person games can be found in Rapoport and Guyer [12].

Acknowledgements

I am grateful to T.C.A. Anant, Gopal Kadekodi, Jean Drèze, Jeroen van der Bergh, Huib Jansen, Anantha Duraiappah, Bob van der Zwaan and two anonymous referees for helpful comments. I am also grateful for constructive comments

from participants in a seminar held at the Economic Institute of the Complutense University, Madrid, 15th June 1999. Remaining errors are mine.

Notes

¹ In the literature we cannot find many approaches to the estimation of games. A few notable ones are [1, 3, 7], which focus mainly on auction data.

² In a more general model the transformation of p_{ij} and q_{ij} into π_i can be assumed to be non-linear as well.

³ Refer to Pongquan [11] for a comparative discussion on the definition of people's participation.

⁴ There are no further limiting assumptions on how function λ_j should behave.

⁵ The actions of the composite second player is defined analogously to portfolio selection. Markowitz [10], for instance, develops six different ways to measure risk as a way to reduce the dimensionality of a choice problem.

⁶ This is possible through a standard procedure of SPSS.

⁷ We assigned the observations into equal groups with a computer program in Turbo Pascal (TP).

⁸ The mean or Euclidean clusters can also be used to realize a division, but this does not guarantee an even distribution of the observations.

⁹ This step is required, because high and low ϑ 's are not necessarily evenly distributed over P and D .

¹⁰ This can be done, for example, with the stepwise-regression method of SPSS. This procedure selects one by one the most significant socio-economic variables, as defined by the user. Only those variables with a user-defined significance level are added into the regression.

¹¹ This can be tested by running the MEANS-T-TEST of SPSS.

¹² Application of the Euclidean Cluster Method requires many more observation for a reliable estimation at the organizational level.

¹³ One quintal is equal to 100 kg.

References

1. Bresnahan, T.F. (1989). Empirical studies of industries with market power. In R. Schmalensee and R.D. Willig (eds.), *Handbook of Industrial Organization, Volume II*. Elsevier Science Publishers, Amsterdam.
2. Coase, R.H. (1992). The institutional structure of production. *American Economic Review*, **82**(4), 713–719.
3. Genesove, D. (1995). Search at wholesale auto auctions. *Quarterly Journal of Economics*, **109**, 23–49.
4. Harman, H.H. (1967). *Modern Factor Analysis*. University of Chicago Press, Chicago, 2nd revised edition.
5. Klein, L.R. (1967). Estimation of interdependent systems in macroeconometrics, *Econometrica*, **37**(2), 171–192.
6. Kloeck, T. and Mennes, L.B. (1960). Simultaneous equation estimation based on principal components of predetermined variables. *Econometrica*, **28**, 45–61.
7. Laffont, J.J. (1997). Game theory and empirical economics: the case of auction data. *European Economic Review*, **41**, 1–35.
8. Lise, W. (1997). An econometric and game theoretic approach to common pool resource management: case studies in rural India. Unpublished Ph.D. thesis, Delhi School of Economics, University of Delhi, India.

9. Lise, W. (1997). Linking indicators of participation in forest management and additional incomes from participating: An estimation procedure for non-cooperative games. Institute of Economic Growth, Delhi, India. Working Paper E/184/97.
10. Markowitz, H.M. (1991). *Portfolio Selection: Efficient Diversification of Investment*. Basil Blackwell, Cambridge, 2nd edn.
11. Pongquan, S. (1992). Participatory development activities at local level: case studies in villages of Central Thailand: an analysis of people's participation in the planned development process using the concept of the 'linking loops'. Ph.D. thesis, Agricultural University Wageningen, The Netherlands.
12. Rapoport, A. and Guyer, M. (1966). A taxonomy of 2×2 games, *General Systems Yearbook*, **11**, 203–214.
13. Tirole, J. (1988). *Theory of Industrial Organisation*. MIT Press, Cambridge.